Data Analyst Project

Profit Analysis

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Introduction

This project analysis the effect of R&D Spending, Administration Spending, and Marketing Spending on the Profit of 50 startups across three states (New York, California, Florida).The goal is to understand which factors most influence profit, predict profit for given input values, visualize insights using Tableau dashboards, provide business recommendations.

Dataset details

The dataset contains 50 Startups Dataset. Attributes are R&D Spending, Administration Spending, Marketing Spending, State (NY, CA, FL) and Profit.

Tools & methodology

Excel

I have done Regression Analysis and find profit prediction in Excel used Multiple Linear Regression with profit as the dependent variable. Independent variables are R&D Spend, Administration, Marketing Spend.

Tableau

For Dashboard creation and visualization

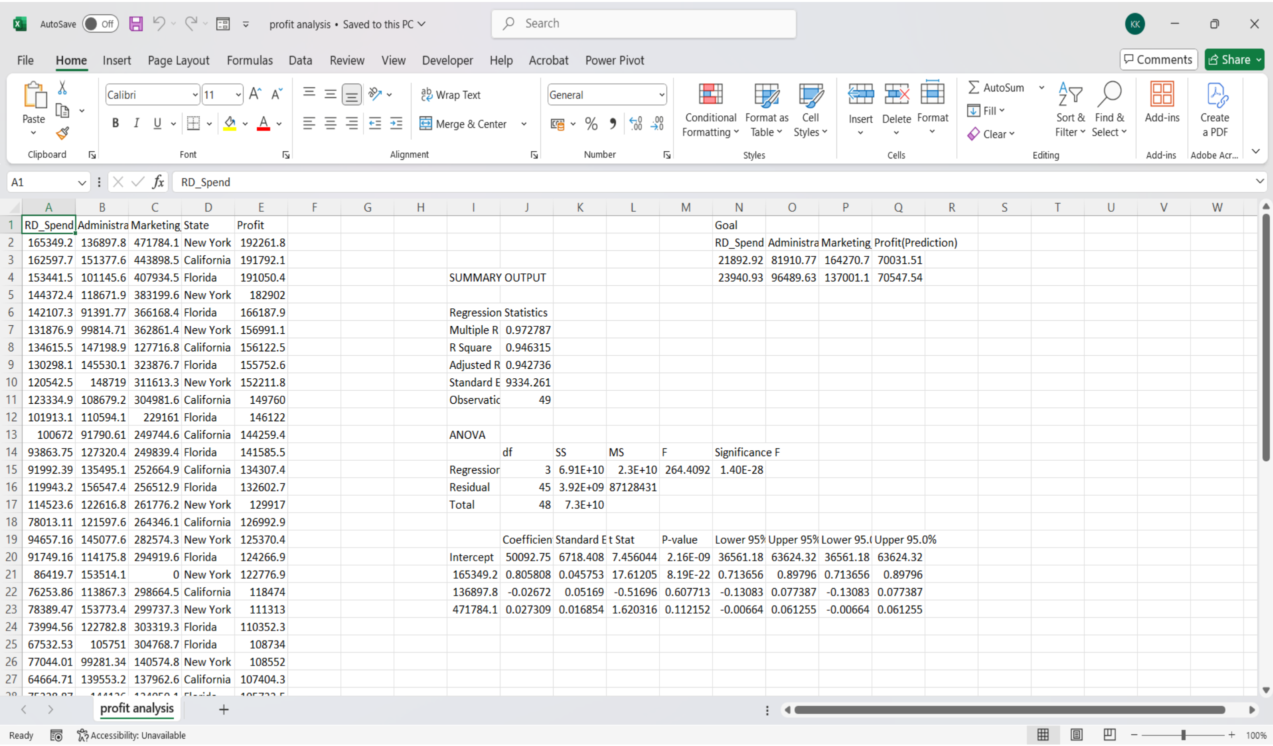
Excel Process

I have done Regression Analysis in Excel used Multiple Linear Regression with profit as the dependent variable.

Formula

Profit = Intercept + (Coeff _ R&D * R&D _ spend) + (Coeff _ Admin * Administration _ spend) + (Coeff _ Marketing * Marketing _ spend).

From regression output, coefficients were obtained to measure impact of each factor. Using the regression formula, profit was predicted.



	A	B	C	D	E
1	RD_Spend	Administra	Marketing	State	Profit
2	165349.2	136897.8	471784.1	New York	192261.8
3	162597.7	151377.6	443898.5	California	191792.1
4	153441.5	101145.6	407934.5	Florida	191050.4
5	144372.4	118671.9	383199.6	New York	182902
6	142107.3	91391.77	366168.4	Florida	166187.9
7	131876.9	99814.71	362861.4	New York	156991.1
8	134615.5	147198.9	127716.8	California	156122.5
9	130298.1	145530.1	323876.7	Florida	155752.6
10	120542.5	148719	311613.3	New York	152211.8
11	123334.9	108679.2	304981.6	California	149760
12	101913.1	110594.1	229161	Florida	146122
13	100672	91790.61	249744.6	California	144259.4
14	93863.75	127320.4	249839.4	Florida	141585.5
15	91992.39	135495.1	252664.9	California	134307.4
16	119943.2	156547.4	256512.9	Florida	132602.7
17	114523.6	122616.8	261776.2	New York	129917
18	78013.11	121597.6	264346.1	California	126992.9
19	94657.16	145077.6	282574.3	New York	125370.4
20	91749.16	114175.8	294919.6	Florida	124266.9
21	86419.7	153514.1	0	New York	122776.9
22	76253.86	113867.3	298664.5	California	118474
23	78389.47	153773.4	299737.3	New York	111313
24	73994.56	122782.8	303319.3	Florida	110352.3
25	67532.53	105751	304768.7	Florida	108734
26	77044.01	99281.34	140574.8	New York	108552
27	64664.71	139553.2	137962.6	California	107404.3

	I	J	K	L	M	N	O	P	Q
1						Goal			
2						RD_Spend	Administra	Marketing	Profit(Prediction)
3						21892.92	81910.77	164270.7	70031.51
4	SUMMARY OUTPUT					23940.93	96489.63	137001.1	70547.54
6	Regression Statistics								
7	Multiple R	0.972787							
8	R Square	0.946315							
9	Adjusted R	0.942736							
10	Standard E	9334.261							
11	Observatic	49							
13	ANOVA								
14		df	SS	MS	F	Significance F			
15	Regressior	3	6.91E+10	2.3E+10	264.4092	1.40E-28			
16	Residual	45	3.92E+09	87128431					
17	Total	48	7.3E+10						
19		Coefficien	Standard E	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.(	Upper 95.0%
20	Intercept	50092.75	6718.408	7.456044	2.16E-09	36561.18	63624.32	36561.18	63624.32
21	165349.2	0.805808	0.045753	17.61205	8.19E-22	0.713656	0.89796	0.713656	0.89796
22	136897.8	-0.02672	0.05169	-0.51696	0.607713	-0.13083	0.077387	-0.13083	0.077387
23	471784.1	0.027309	0.016854	1.620316	0.112152	-0.00664	0.061255	-0.00664	0.061255

Dashboard in Tableau

Created dashboards to analysis

Profit by State → Bar Chart.

R&D Spend vs Profit → Scatter Plot with Trend Line.

Marketing Spend vs Profit → Scatter Plot with Trend Line.

Profit Contribution by State → Treemap.

KPI Cards → Total Profit, Average Profit, Maximum profit.

Insights

- R&D Spend has the strongest influence on Profit (high coefficient).
- Administration Spend shows little to no effect on Profit.
- Marketing Spend has moderate influence but less than R&D.
- State-wise Analysis: Certain states (e.g., California) show higher profit averages compared to others.
- KPI Summary:

  Total Profit : 5,600,632

  Average Profit : 112,013

  Highest Profit observed : 192,262

Recommendations

- Increase investment in R&D activities since it drives maximum profit growth.
- Optimize Marketing strategies to improve efficiency of spend.
- Minimize Administration overhead costs, as they have negligible contribution to profit.
- Focus expansion in states where profit contribution is consistently higher.

Conclusion:

- The project successfully demonstrated how regression analysis and data visualization can help a company understand key profit drivers. Insights confirm that R&D spending is the most critical factor for profit growth, while administration costs can be controlled.

- By following the recommendations, startups can allocate resources more effectively and maximize returns.